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ORIGINAL PAPER

# Improving Compliance in Primary School Students with Autism Spectrum Disorder

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## Abstract

Complying with common instructions is considered an important skill, critical to school success; however, students with autism spectrum disorder (ASD) often exhibit low levels of compliance creating barriers to their inclusion in regular general education school settings. While self-management interventions have the potential to address compliance issues, there has been little research investigating their effectiveness in regular education school settings that include young children with ASD. Accordingly, the present study examined the effects of a self-management intervention for two 8-year-old boys with ASD and attention-deficit hyperactivity disorder. A multiple-baseline across settings design was used. Results indicated that the intervention was associated with increased rates of compliance and concomitant increases in on-task behavior for both participants within their respective classroom. Effects were maintained at follow-up, and social validity ratings suggested that the intervention was highly acceptable for both the students and their teachers. This study contributes to the knowledge base on effective and feasible interventions to support the inclusion of children with ASD in general education settings.

**Keywords** Compliance · Self-management · Autism · Multiple-baseline experimental design · On-task behavior

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## Introduction

Compliance can be thought of as a skill that develops throughout childhood and for the purposes of this study is defined as "acting in accordance with a directive to engage in or to stop engaging in a behaviour" (Owen et al. 2012, p. 364). Low levels of compliance can be accompanied by behaviors of concern often referred to in the literature as *non-compliant* behaviors (such as refusal or defiance; Kochanska and Aksan 1995). This can be confusing as the prefix suggests an absence of a behavior or skill, yet often the term non-compliance is used to indicate behavioral excesses. In this report, we will refer to compliance as a skill and non-compliance as its absence.

Of significance, non-compliance is a problem commonly reported by those who work with children with autism spectrum disorder (ASD; Soto-Chodiman et al. 2012; Van Bourgondien 1993). Interestingly, some of the defining characteristics of ASD, such as deficits in social and communication skills, may make it difficult for such children to comply with common instructions. Within school environments, persistent low levels of compliance can adversely impact a student's ability to develop appropriate social and academic skills (Austin and Agar 2005). Further, compliance problems can cause significant distress for teachers (Aloe et al. 2014) and limit educational opportunities for other students in the class (de Martini-Scully et al. 2000).

A number of intervention programs have been developed to address non-compliance in young children. Three commonly utilized strategies to enhance compliance are the high-probability command sequence (HPCS; Nevin 1996), errorless compliance training (ECT; Ducharme 1996), and effective instruction delivery (EID; Ford et al. 2001). Despite improved rates of compliance being reported in some instances, research on each has yielded mixed results and identified clear limitations to their application in applied settings, such as schools (Banda et al. 2003; Ducharme and Shecter 2011; Lui et al. 2014). Common limitations include a requirement for skillful procedural implementation by others (e.g., parents or teachers) and low rates of treatment fidelity.

Given the limitations of these strategies in academic settings, further research exploring treatment alternatives appears warranted. Ideally, such strategies should be characterized by ease of implementation, with reduced reliance on prompting by adults. Preferably, these characteristics would make interventions attractive for teachers allowing them more time for academic instruction and requiring less for behavior management.

One strategy that may satisfy these criteria is self-management. Self-management has been defined as "… the personal application of behavior-change tactics that produces a desired change in behavior" (Cooper et al. 2007, p. 586). A well-established, evidence-based procedure (National Autism Center 2009), including in primary schools (Briesch et al. 2019; Busacca et al. 2015; Carr et al. 2014; Moore et al. 2013), self-management commonly includes strategies such as self-monitoring, self-evaluation, and self-reinforcement. Self-monitoring is a strategy whereby a person systematically observes his/her behavior and records the occurrence or non-occurrence of the target behavior, self-evaluation involves creating

a behavioral goal and evaluating one's own performance against a predetermined goal, and self-reinforcement includes the delivery of a reward contingent upon meeting a predetermined goal. These strategies can be used in isolation, but often they are used in combination with at least one other strategy (Briesch et al. 2019; Cooper et al. 2007). Self-management is considered to be a pivotal skill that may generate widespread behavioral gains (Koegel et al. 1999). Self-management interventions have been reported to be effective in schools (Busacca et al. 2015), in improving both on-task behavior and academic performance of children with ADHD (Slattery et al. 2016) and children with ASD (Carr et al. 2014). Indeed, a meta-analysis conducted by Lee et al. (2007) reported self-management effective in increasing the frequency of appropriate behavior of students with ASD, in a variety of settings and conditions. Further, Wilkinson (2008) reported that students who acquired self-management skills displayed enhanced academic and social skills as a by-product of being able to manage their own behavior more effectively. Scruggs and Mastropieri (1998), in a review article describing the implementation of self-management interventions for children with ASD in general education schools, cites evidence for increased independence and competence as additional benefits of self-management.

There is, however, limited research investigating the utility of self-management interventions to increase compliance rates in individuals with ASD, although the studies that have been conducted demonstrated potential for further investigation. For example, Wilkinson (2008) used a self-management procedure to increase compliance rates in a 9-year-old student with Asperger's syndrome in a general education school setting while Lui et al. (2014) investigated the effectiveness of self-management strategies in improving compliance with parental requests in the home setting. In the Lui et al. study, the young boys diagnosed with either Asperger's syndrome or ASD all showed marked improvements in compliance and a reduction in problem behavior on implementation of the procedure with their parents reporting the intervention to be both acceptable and easy to implement. This study also provided support for the notion that self-management was a pivotal skill, since all the children displayed improvements in untargeted problem behaviors and their parents reported increased independent functioning among their children.

Although self-management is considered an effective intervention, gaps in the literature have been identified. Few studies have been conducted with younger children with ASD in general education settings (Aljadeff-Abergel et al. 2015). For example, Busacca et al. (2015) in their review of 16 studies found a paucity of self-management research conducted on students in Grades 1 and 2, with a particular lack of quality studies investigating the effectiveness of self-management interventions for students with ASD, learning disabilities, and emotional and/or behavioral disorders. Further, a lack of research investigating the effectiveness of self-management in increasing compliance among children with ASD has been noted (Aljadeff-Abergel et al. 2015; Busacca et al. 2015). At the same time, the educational needs of these students are increasingly having to be met in general education classrooms by teachers who often feel ill prepared to meet their needs (de Boer et al. 2011). There is therefore a need for effective procedures that are easy to implement in schools and that do not place excessive demands on teachers.

The purpose of this study, therefore, was to investigate the effectiveness of a self-management intervention for two students diagnosed with ASD and ADHD in a Grade 2 regular class setting. It was anticipated that the implementation of self-management would result in improved compliance, along with an expectation that there would be a concomitant improvement in on-task behavior. It was also predicted that compliance and on-task behavior would be maintained following fading of the intervention and that both the students and the teachers would consider the self-management intervention socially valid.

## Methods

Ethics approval was obtained from the University Research Ethics Committee before recruitment took place. The school principals also gave permission to conduct the study in their schools, and the participants, their parents, and teachers gave informed consent for participating in the project prior to data collection.

### Recruitment and Selection

Information about the research project was advertised on a state-level ASD peak body research registry. Interested parents contacted the researchers before undergoing a brief phone interview to screen for their children's suitability for the project. Eligibility criteria for participation were: The child has a diagnosis of autism spectrum disorder, is enrolled in the general education system, either in a private or catholic school, and in Grade 1 or 2. The teacher of the child has previously told parents that they have difficulty getting the child to follow instructions at school.

After consent and permission were obtained, more detailed participant information was gained in a face-to-face interview with the teachers.

### Participants

The participants in this study were two 8-year-old boys. The first participant was a Grade 2 student in a private school in Melbourne. According to his parents, Sean (pseudonym) was diagnosed with Asperger's disorder at 4.5 years old and, subsequently, also with attention-deficit/hyperactivity disorder (ADHD). Since the diagnosis, Sean had been receiving therapy based on the principles of applied behavior analysis (ABA). Sean was also on medication (risperidone and fluoxetine) at the point when his parents approached the researchers. Despite the interventions received, Sean's mother reported that he had difficulty following instructions at school. At school, Sean received additional individual in-class support from his ABA therapist (once a week, 40 min), parent volunteers (once a week, 60 min), and an integration aide (three times per week, 80 min each session on average) throughout the project duration. These additional support personnel were mostly present during English classes, when Sean tended to display non-compliant and more off-task behavior.

The second participant was a Grade 2 student at a different private school in Melbourne. John (pseudonym) was diagnosed with ASD at 6 years old and subsequently also with ADHD. According to the psychological, medical, and speech pathology reports provided by his parents, he had a complex medical history including prematurity, global development delay, oro-motor (non-speech) and sensory related difficulties, along with pragmatic language disorder. Since diagnosis, John had been receiving extensive behavior therapy, social skills training, and speech therapy from various agencies. He was also on medication at the point when his parents approached the researchers. Despite the interventions and medications received, John's mother reported that he had difficulty following instructions at school. At school, John received in-class support from an integration aide before and during the project. The integration aide was mostly present during reading and writing classes where John was reported to have difficulty attending and staying on task.

## Settings

Most observations and the intervention were conducted in the participants' classrooms. Sean's class consisted of 17 students. Classes generally were of 40-min duration, though occasionally activities would stretch beyond 40 min if multiple consecutive slots were allocated for a subject. A class typically began with whole class instruction by the teacher, with students seated on the floor at the front of the room. This was followed by independent or group work at the students' desks. The use of technology such as smartboards, iPads, and computers was common in the classroom. Due to a term change, there were slight changes in the class schedule and support structure over the course of the study. Three subject areas were identified for intervention: Math, English, and English support classes. English support classes were phonics lessons in a smaller class of eight students in a different classroom with a different teacher.

John was in a class of 24 students. Classes were of 50-min duration, and occasionally joint activities were held across two classes that stretched beyond 50 min if consecutive slots were allocated for a subject. A class typically began with whole class instruction on the floor where the teacher explained the task by giving an example and asking questions. This was followed by students working on the task at their desks. Following preliminary observations and consultations with the class teacher, three subjects were targeted for intervention: reading, writing, and numeracy classes.

## Materials

### Criteria Checklist

A checklist consisting of five questions was used as a semi-structured interview protocol when interested parents initially contacted the researcher for eligibility screening.

### Event Recording form for Compliance

Event recording forms were created to record observed instances of compliance and non-compliance with instructions in each session.

### Momentary Time Sample Recording form for On-Task Behavior

Momentary time sampling forms were created to record on-task behavior in each session. The form also allowed the researcher to collect the data on on-task behavior of two other students in the classroom for a normative comparison of on-task behavior.

### Story: Following Instructions

A 52-word story was created, based on Lui et al. (2014) to describe what compliance is. The story explains what instructions are (instructions are about things that the teachers want me to do) and why it is good to follow instructions (e.g., When I follow instructions, it makes me learn better.)

### Role-Play Situations

Between 15 and 20 role-play scripts were used during the teaching phase. They were crafted based on the information gathered during preliminary observations and included teacher requests such as "Take out your book" and "Let's tidy up the table."

### Self-Monitoring Sheet

Recording sheets were developed for the participants to self-record incidents of compliance. For John, this was a reusable laminate in accordance with the school policy on ecological sustainability. The recording sheets consisted of 20 blank boxes (with hook and loop fasteners on the laminated version) where participants were instructed to paste a smiley face for each instance of compliance. The sheets also included spaces for the agreed goal and reward.

### Treatment Fidelity Checklist

The checklist was created to measure the extent to which the five components of the intervention package were delivered as planned during each intervention session. (All the above materials are available from the first author on request.)

### Behavior Intervention Rating Scale (BIRS)

The BIRS was created by Elliott and Treuting (1991) to assess teachers' acceptability and perceived effectiveness ratings of educational interventions. It consists of 24 items with a 6-point Likert scale. This form was used for Sean, while for John the adapted version (BIRS-A) was used. The adapted version includes minor wording

changes so as to be able to use the scale as a pre- and post-measure (i.e., Item 1 "This would be…" = "This will be….." in the adapted (pre-) version).

#### Children's Intervention Rating Profile-Adapted Version (CIRP-A)

An adapted version of CIRP (Turco and Elliott 1986) was developed commensurate with the participants' age. Although CIRP-A is similar to the original version that consists of seven items and a 6-point Likert scale, the adapted version included simplified language and replacement of the Likert scale with emoji.

### Independent and Dependent Variables

#### Independent Variable

The independent variable was a self-management intervention package consisting of the following components: goal setting, discrimination teaching, self-monitoring, self-recording, and reinforcement.

#### Dependent Variable

The level of compliance to instructions was the primary dependent variable. Compliance was operationally defined as the initiation of a requested response within 10 secs of the teacher's request. It was expressed as a percentage of the number of instructions delivered within the observed period. It was calculated by dividing the number of instructions complied with by the total number of instructions delivered and multiplying by 100.

The concomitant measure was on-task behavior, operationally defined as behaviors that the student was expected to be engaged in at a particular moment in time. These included behaviors such as sitting in seat, paying attention to the teacher when the teacher is instructing (e.g., listening, eyes and face forward), and working on the assignment. On-task behavior was calculated by dividing the number of observed intervals on-task by the total number of observed intervals and multiplied by 100.

### Experimental Design

A multiple-baseline across three settings (reading, writing, and numeracy classes for John; and Math, English, and English support classes for Sean) design was used to evaluate the functional relationship between the self-management intervention and observed changes in the participants' behavior. In this design, the baseline phase was initiated simultaneously in all the three settings. Initiation of the intervention phase was time-lagged sequentially across the subject areas.

## Data Collection

Data were collected over the course of 3 months, three to four times a week for each class.

### Observation Procedure

Event recording was used to document incidents of the participants' compliance and non-compliance for the entire duration of each observation, whereas data on on-task behavior were gathered using momentary time sampling of 10-s intervals for a discrete activity typically the first activity with clear starting and ending points that occurred during the observation. To make a normative comparison for on-task behavior, data from two peers who happened to be near the target student were also collected in each setting at the same time. Rates of on-task behavior were calculated by dividing the number of intervals observed with on-task behavior by the total number of observation intervals, multiplied by 100%. The peers' on-task behavior data were calculated in the same manner and then averaged to provide a single estimate for comparison.

### Inter-observer Agreement

Inter-observer agreement was assessed for 22–33% of the observation sessions across all classes in both baseline and intervention phases, to measure the reliability of the observations. The observations were taken independently but simultaneously. Percent agreement was calculated using the interval by interval method and dividing the number of agreements by the number of agreements plus disagreements, and then multiplied the value by 100%. Agreements were defined as instances in which both observers' recordings matched interval by interval.

For John, the mean inter-observer agreement for compliance was 99% (range 80–100%), and on-task behavior mean was 96% (range 77–100%), and for Sean the mean inter-observer agreement for compliance was 97% (91–100%), with the on-task behavior mean at 86% (64–98%). According to Hartmann, Barrios, and Wood (2004), minimum acceptable values of an overall inter-observer agreement should range between 80 and 90%. Thus, overall satisfactory levels of inter-observer agreements were obtained.

### Treatment Fidelity

Treatment fidelity data were collected in each observation session during the intervention and fading phases with the treatment fidelity checklist. Treatment fidelity was measured to determine the extent to which the implementation of the intervention adhered to planned procedures (Cooper et al. 2007). Adherence to the planned procedures was calculated by dividing the number of steps completed correctly by the total number of steps, with a calculated mean of 100% fidelity, meaning all steps of the self-management procedure were followed correctly during all intervention and fading sessions.

## Procedures

### Preliminary Observations

Preliminary observations were conducted 2 days before the collection of baseline data to select appropriate subject areas for intervention, to have a better understanding of the participants' behavior, to customize the teaching materials, and to allow the students and teachers to become accustomed to the presence of the researchers in their classrooms.

### Baseline Phase

During baseline, unobtrusive observations were conducted to collect participants' behavior data during typical daily routines in the respective classrooms and sessions. Researchers positioned themselves at the rear of the classrooms, ignoring any attempts by the students to interact or communicate with them. If a student attempted to initiate interactions, the researcher pretended to be preoccupied, to minimize the effect of the researcher's presence in the classroom.

Following the completion of baseline, the researchers conducted a preference assessment for John by way of interviews with him and his teacher to determine a list of preferred items and activities that were also acceptable and feasible. Sean volunteered his preferences informally.

### Intervention Phase

The intervention phase consisted of two parts: training and self-monitoring.

**Part 1: Training** The aim of this was to teach participants to discriminate between compliance and non-compliance and to train them to implement the self-management procedures. Participants attended two 40-min training sessions across 2 days. The training consisted of a series of steps, including: outlining the purpose of the study, describing compliance as it related to the study, and using role-play to teach the difference between compliance and non-compliance. Then, we coached participants in the use of the self-management procedure, by having them describe the procedure, training them to use the self-monitoring tool, and providing opportunities to practice the procedure. Training was completed once participants achieved at least 80% accuracy in self-recording in two consecutive model activities.

**Part 2: Self-Monitoring** Five minutes prior to the beginning of the target observation class, the self-management goal and reward were set for the session in consultation with participants and the teacher if appropriate. Participants were then handed the self-monitoring sheet, on a clipboard. John also received stickers, while Sean also received a pencil to self-record. During the class, a researcher sat behind the participants to prompt them to self-record through either verbal or

nonverbal means (e.g., shoulder taps) if they did not self-record within 5 secs after performing a compliant behavior. Non-compliance was ignored. At the end of the class, the researcher collected the self-monitoring sheets from the participants and reviewed them. If participants achieved the self-management goal for the class, rewards were administered at the next convenient time.

### Fading Phase

The fading procedure was based on that described in Lui et al. (2014). Fading involved the gradual withdrawal of the self-management procedure in each setting once a steady high rate of compliance was achieved. For John, this was accomplished by increasing the requirements to receive a reward. Next, the frequency of using the self-monitoring tool was decreased until finally it was no longer used. Instead of providing John with self-monitoring sheets, the researcher instructed John to mentally tally the number of compliant acts during the class. For Sean, the first step was omitted and fading began with the intermittent use of the self-monitoring sheet as described above. All other steps (i.e., setting of goals and rewards, review of performance, and administration of rewards) remained unchanged. In addition, for both John and Sean, there was a shift of rewards from tangible (e.g., free computer time after the class) to social reinforcements (e.g., verbal praise by the teacher).

### Follow-up

Follow-up observation sessions were conducted after the last observation of the fading condition: 7–10 days later for John and 1 week later for Sean. The follow-up sessions were conducted under baseline conditions.

### Social Validity

Social validity in terms of the social significance of the goals, outcomes, and procedures of self-management (Wolf 1978) was assessed by asking the teachers and participants to respond to the BIRS (BIRS-A for John) and the CIRP-A in baseline and following the fading phase. Anecdotal social validity information was also collected from the participants and their teachers throughout the study.

### Data Analysis

The data were graphed and analyzed using visual analysis, the most common data analytical technique in single-case experimental design (Busk and Marascuilo 2015). The effectiveness of the intervention was computed using the percentage of non-overlapping data (PND) metric, this being the percent of intervention data points that surpassed the highest baseline data point (Scruggs et al. 1987; see also Scruggs and Mastropieri 1998). Although as yet there is no consensus regarding

the most appropriate procedure for estimating the effectiveness of interventions in single-subject design studies (Parker et al. 2007), a recent review concluded that the PND metric results in coherent and valid estimates of effectiveness of intervention in single-subject research (Carr et al. 2015). According to the standard established by Scruggs et al. (1987), PND scores over 90% are interpreted as *very effective*, 70% to 90% as *effective*, 50% to 70% as *questionable*, and below 50% as *ineffective* treatments.

## Results

The effectiveness of the self-management intervention was evaluated using visual analysis. For within-condition analysis for both compliance and on-task behavior, three elements were examined: level, trend, and variability of data. As per the guidelines by Cooper et al. (2007), the median was used instead of the mean in analyzing compliance due to the presence of several extreme values in the data set. Analysis of level, trend, and variability indicated that participants demonstrated increases in compliance and on-task behavior from baseline to self-management intervention, and the increases were maintained when the intervention was faded.

The percentage of Sean and John's compliance is presented graphically in Figs. 1 and 2, respectively. Visual analysis indicates that the percentage of compliance increased in both participants in association with the staggered implementation of the intervention across all settings, and the increases maintained when the intervention was faded.

### Participant 1: Sean

#### Math

In the baseline phase, Sean showed a moderate, stable level of compliance with an ascending trend (*Mdn* = 51.3%, range 44.4–55.6%). Upon the introduction of the self-monitoring intervention, his compliance increased to a moderate-to-high level with a median of 76.5% (69.2–84.6%), with a stable and slightly descending trend. During the fading phase, his compliance remained at moderate-to-high levels with a median compliance rate of 76.0% (72.2–80.9%) with a stable, descending trend. His high compliance was maintained 1 week later at 90.0%.

#### English

Sean's baseline compliance was at moderate-to-low levels with a steadily declining trend (*Mdn* = 38.7%, range 22.7–42.9%). During the intervention phase, his compliance improved to a median of 64.0% (57.1–70.8%), with an initially moderate-to-high, ascending level of compliance followed by a high, stable level of compliance. Due to the term change midway through the project, there was a change in the support schedule for Sean, such that he had adult support in every English class in the

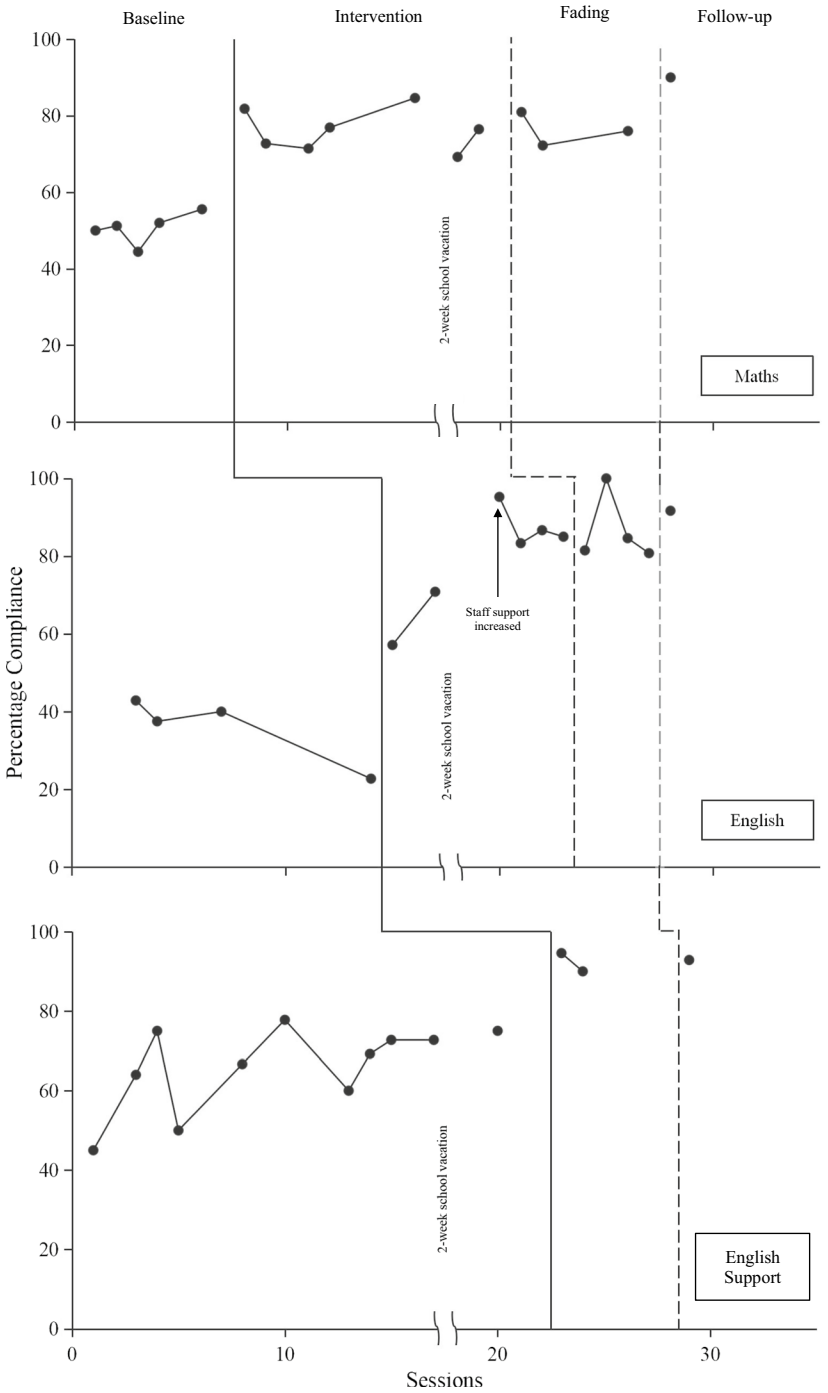



**Fig. 1** Sean's percentage of compliance across settings

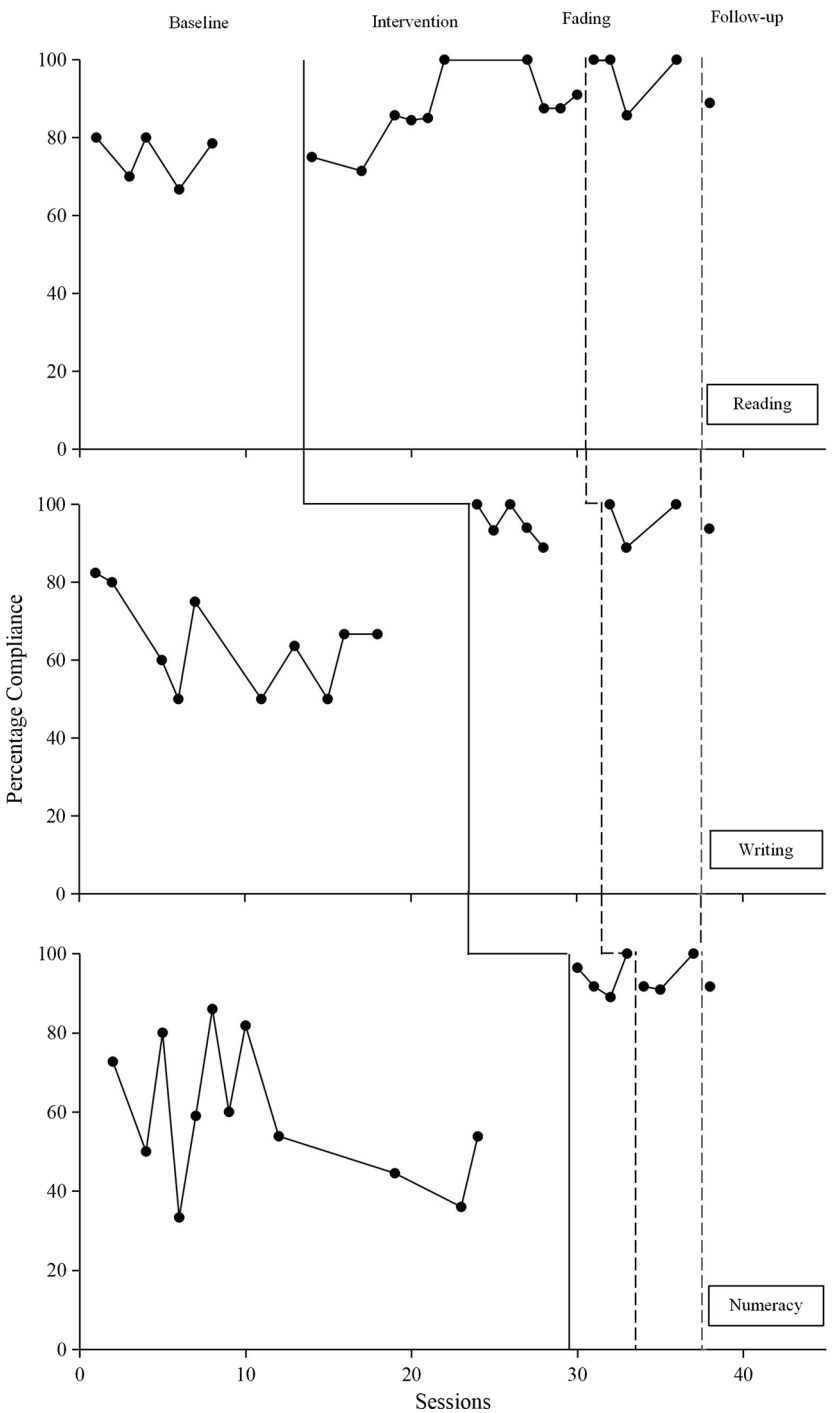



**Fig. 2** John's percentage of compliance across settings

later part of the term. When both adult support and self-monitoring were in place, Sean's compliance increased further (*Mdn* = 85.8%, range 83.3–95.2%), with a high, relatively stable descending trend. Although his compliance remained at a high level, when the self-monitoring tool was withdrawn, Sean's compliance showed an unstable, descending trend with a median percentage of 83.0% (80.8% to 100%). His rate of compliance at the 1-week follow-up was maintained at 91.7%.

#### English Support

Sean displayed a moderate-to-high, variable level of compliance during the baseline observations (*Mdn* = 69.2%, range 45.0–77.8%). Upon the introduction of the intervention, Sean's compliance increased to a high level with a median of 92.3% (90.0% to 94.6%), with a stable trend. However, due to school events and changes in the class schedule, English support classes became very tentative; consequently, the intervention was suspended due to the unpredictable schedule of the classes. Nevertheless, Sean's compliance remained high at 92.9% at follow-up.

### Participant 2: John

#### Reading

In baseline, John displayed a moderate level of compliance with a relatively stable trend with a median of 78.5% (range 66.7–80.0%). Upon the implementation of self-management, his level of compliance improved gradually to a median of 86.6% (range 71.4–100%), with a slight variable trend evident. During fading, John's level of compliance was maintained at a high level with a median of 100% (range 85.7–100%) with maximum possible scores over the first two and the last fading sessions. At 10-day follow-up, John maintained a high level of compliance at 88.9%.

#### Writing

John displayed variable moderate baseline levels of compliance during the baseline phase with a median of 65.2% (range 50–82.4%). No clear trend was visible. When self-management was implemented, there was an abrupt increase in his level of compliance with no overlap with baseline data. John scored a high level of compliance (*Mdn* = 94.0%, range 88.9–100%), with a stable trend. During fading, John maintained a high level of compliance with maximum possible scores on the first and the last fading sessions with a median percentage of 100% (range 88.9–100%). Similar to the reading class, John maintained high levels of compliance at 93.8% at follow-up.

#### Numeracy

During the baseline phase, John displayed a relatively moderate level of compliance with a median of 56.4% (range 33.3–86.0%) in the numeracy class. There was no

detectable trend due to high variability. Implementation of self-management led to a marked increase in the level of compliance (*Mdn* = 94.1%; range 89.0–100%) with a slightly descending trend but with no overlap with baseline data. Similar to the reading and writing classes, when the self-monitoring tool was gradually withdrawn and eventually removed, John's level of compliance maintained with a median of 91.7% (range 90.9–100%). His level of compliance at 10-day follow-up maintained at 91.7%.

### Effectiveness of Intervention

For Sean, the PND scores of the intervention in all settings (i.e., Math, English, and English support) were 100%. According to Scruggs et al. (1987) standard, interventions with PND scores over 90% are considered very effective. Thus, it can be concluded that for Sean the introduction of the self-management intervention was very effective in increasing compliance in all settings.

For John, the PND score of the intervention during reading class was 80.0%, which indicates that the intervention was effective by the Scruggs et al. (1987)standards, in increasing John's level of compliance. The implementation of self-management was very effective in increasing compliance during writing and numeracy classes (PND scores for both = 100%). Calculated across the three intervention settings, the average PND effect size was 93.3%, suggesting that the introduction of self-management was very effective in increasing John's compliance.

Figures 3 and 4 show the percentages of on-task behavior for Sean and John, and their peers, respectively.

### Participant 1: Sean

**Math** The level of Sean's and his peers' on-task behavior showed obvious variation across the days. Sean consistently displayed less on-task behavior (*Mdn* = 48.0%, range 10.0–65.0%) than his peers (*Mdn* = 88.0%, range 75.0–97.5%) during the baseline observations, with a median percentage difference of −40.0% (range −65.0 to −30.0%). When self-management was introduced, the gap between Sean (*Mdn* = 66.7%, range 27.3–100%) and his peers (*Mdn* = 77.3%, range 55.9–87.2%) reduced, with a median percentage difference of −3.6% (range −40.9% to 22.7%). There were two occasions (Days 12 and 16) when Sean was more on-task than his peers. A possible explanation for the outlier data points on Day 19, for both Sean and his peers is that there was a school event immediately after the class. During the fading phase, Sean was consistently more on-task (*Mdn* = 60%, range 50–66.7%) compared to his peers (*Mdn* = 45.8%, range 36.7–56.7%), with a median percentage difference of 10% (range 4.2–23.3%).

**English** Similar to the data from Math, there was a consistent difference in levels of on-task behavior between Sean (*Mdn* = 47.9%, range 23.1–56.3%) and his peers (*Mdn* = 75%, range 53.9–90%) during the baseline phase, with a median percentage difference of −30% (range −40 to −18.8%). When self-management was

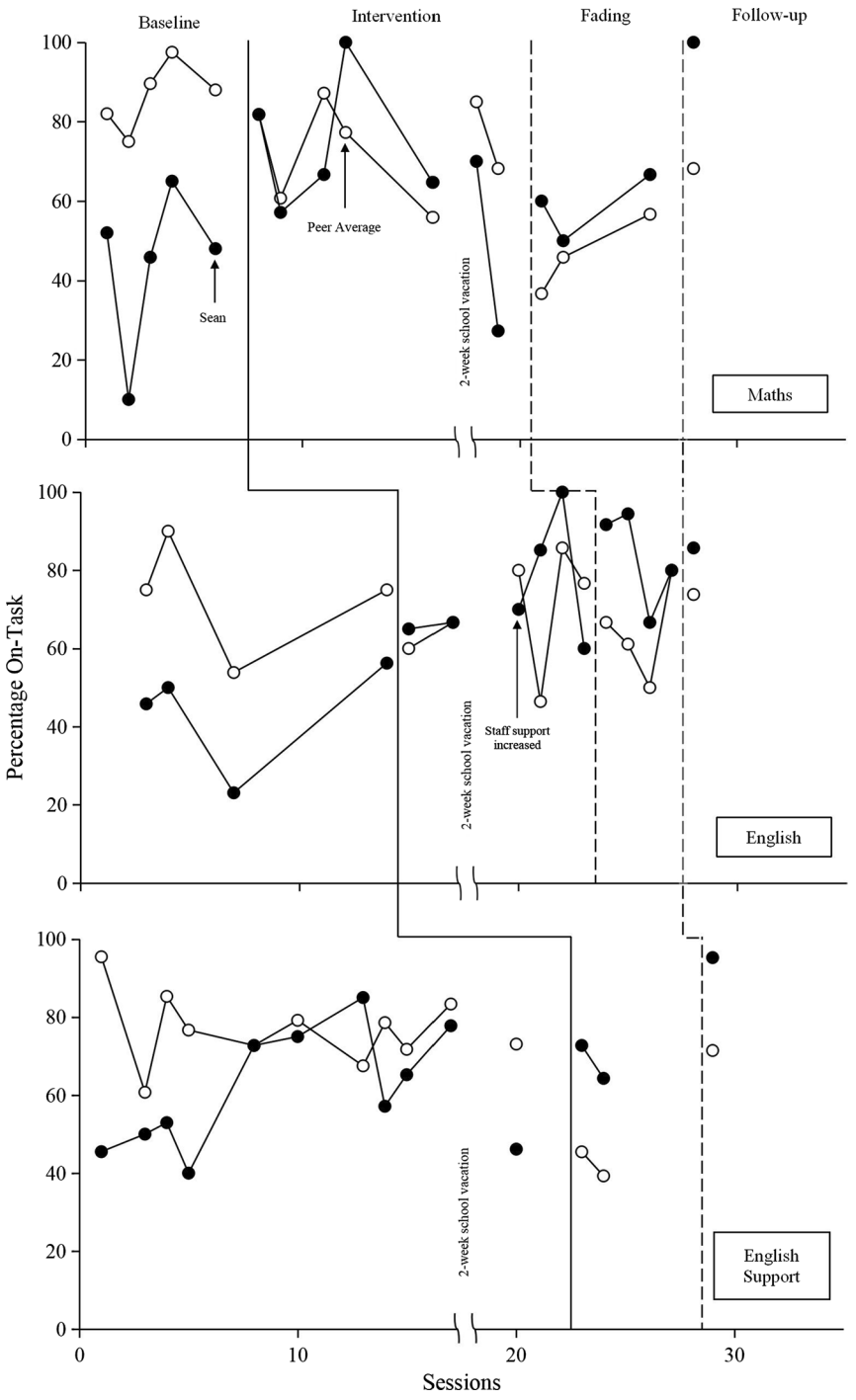



**Fig. 3** Percentages of Sean's and his peers' on-task behavior across settings

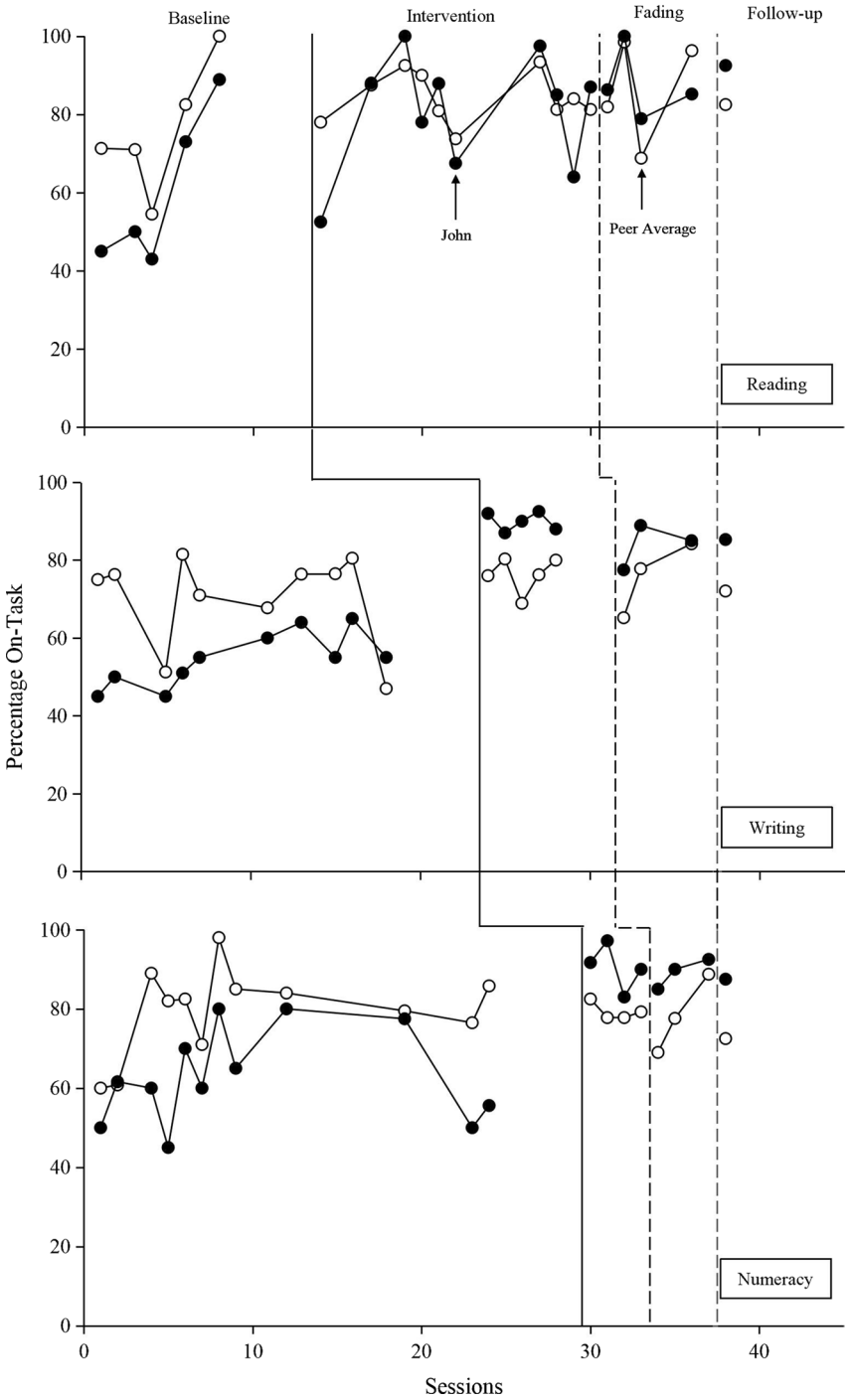



**Fig. 4** Percentages of John's and his peers' on-task behavior across settings

introduced, Sean's on-task behavior (*Mdn* = 68.3%, range 60–100%) became comparable to that of his peers (*Mdn* = 71.7%, range 46.4–85.7%), with a median percentage difference of 2.5% (range − 16.7 to 38.7%). With adult support, his on-task behavior continued to be comparable, or better than his peers. When the use of the self-monitoring tool was faded out, Sean's on-task behavior (*Mdn* = 85.8%, range 66.7–94.4%) remained comparable to or higher than that of his peers (*Mdn* = 63.9%, range 50–80%), with a median percentage difference of 20.8% (range 0–33.3%).

**English Support** Sean's (*Mdn* = 57.1%, range 40–85%) and his peers' (*Mdn* = 76.7%, range 60.7–95.5%) on-task behavior was very variable during the baseline phase with a median percentage difference of − 10.7% (range − 50 to 17.5%). Sean was less on-task than his peers before Day 5, the gap narrowed from Day 8 onwards, making his on-task behavior comparable to his peers. This coincided with the introduction of the self-management intervention during the Math class on the same day. It appeared that there could be some generalization effect across settings with the use of the self-monitoring tool. When self-management was introduced, Sean (*Mdn* = 68.5%, range 64.3–72.7%) was consistently more on-task than his peers (*Mdn* = 42.4%, range 39.3–45.5%). The median percentage difference in the intervention phase was 26.1% (range 25–27.3%). However, the intervention was suspended due to the unpredictable schedule of the classes.

### Participant 2: John

**Reading** During the Baseline phase, John consistently displayed lower levels of on-task behavior (*Mdn* = 50%, range 43–88.9%) than his peers (*Mdn* = 71.3%, range 54.5–100%) with a median percentage difference of − 12% (range − 26.3% to − 9.5%) and with an increasing trend evident for both John and his peers With the introduction of self-management, the gap narrowed, making John's on-task behavior (*Mdn* = 86%, range 52.5–100%) comparable to his peers (*Mdn* = 82.6%, range 73.8–93.4%) with a median percentage difference of 2.1% (range − 25.5–7.5%). When the use of the self-monitoring tool was faded out, John's on-task behavior remained comparable (*Mdn* = 85.8%, range 78.9–100%) to his peers (*Mdn* = 89.1%, range 68.8–98.5%), with a median percentage difference of − 4.2% (range − 11.1% to 10.1%).

**Writing** Similar to the data from reading, both John and his peers showed considerable variability in on-task behavior during the baseline phase. While John displayed a relatively stable moderate level of on-task behavior (*Mdn* = 55%, range 45–65%), his peers displayed a moderate-to-high level of on-task behavior with some variability (*Mdn* = 75.7%, range 47–81.5%). There was a median percentage difference of − 15.8% (range − 30.5% to 8%). Although overall John displayed less on-task behavior than his peers, there was one occasion (session 18) where he was more on-task than his peers. With the introduction of self-management, John's on-task behavior improved and stabilized (*Mdn* = 90%, range 87–92.5%), and he was consistently more on-task than his peers (*Mdn* = 76.3%, range 68.9–80.3%),

with a median percentage difference of 16% (range 6.7–21.1%). When the use of the self-monitoring tool was faded out, John's on-task behavior (*Mdn* = 85%, range 77.5–88.9%) remained comparable to, or higher than his peers' on-task behavior (*Mdn* = 77.8%, range 65.2–84.2%), with a median percentage difference of 11.1% (range 0.8–12.3%).

**Numeracy** Similar to the data from both reading and writing, the level of John's and his peers' on-task behavior showed considerable variation during the baseline observations. John was consistently less on-task (*Mdn* = 61.6%, range 45–80%) than his peers (*Mdn* = 82.5%, range 59.5–98%) with a median percentage difference of − 18% (range − 37% to 4.5%). When self-management was introduced, John displayed a high level of on-task behavior with reduced in-phase variability (*Mdn* = 90.9%, range 83–97.2%). While his peers displayed a stable and high level of on-task behavior (*Mdn* = 78.5%, range 77.8–82.5%), John stayed consistently more on-task than his peers. The median percentage difference in the intervention phase was 10% (range 5.2–19.4%). During the fading phase, John maintained higher on-task behavior (*Mdn* = 90%, range 85–92.5%) than his peers (*Mdn* = 77.6%, range 69–88.8%), with a median percentage difference of 12.4% (range 3.8–16%).

The prediction that compliance and on-task behavior would be maintained following fading of the intervention was met. For Sean, in math his high compliance was maintained 1 week later at 90.0%, while in English his rate of compliance at the 1-week follow-up was maintained at 91.7% and in English support after a 1-week follow-up, he remained more on-task than his peers, with a percentage difference of 23.8%.

For John, in the reading task at 10-day follow-up, he remained more on-task than his peers, with a percentage difference of 10%. The intervention effect in the writing activity was maintained at 10-day follow-up with a percentage difference of 13.2%, while in numeracy, similar to the other two settings, John continued to engage in more on-task behavior than his peers at 10-day follow-up with a percentage difference of 15%.

## Social Validity

### Participant 1: Sean

Sean expressed an overall favorable view of the study before and after the intervention. He continued to like the intervention (Item 6) and believed that the intervention would help him do better in school (Item 7). Sean found the intervention to be difficult, as indicated by a 3-point change in his ratings on pre- and post-measures on Item 3 "The method is too hard on me."

Despite the already-positive ratings on the pre-measure, his teacher's ratings on the post-measure increased by 1–2 points for 17 out of the 24 items on the rating scale. There was a change in polarity of the teacher's perception of the acceptability of intervention (Items 3 and 4), lasting impact of intervention (Items 17 and 20),

magnitude (Item 18), and rate of behavioral improvement (Items 16 and 19), with more favorable ratings on the post-measure.

#### Participant 2: John

Overall, there was evidence that the intervention procedure was socially acceptable for both John and his teacher. John held a favorable view of the study before and after the intervention. He continued to like the intervention (Item 6) and believed that the intervention would help him do better in school (Item 7). At the start of the study, John was concerned that the self-management procedure would cause some problems with his peers as indicated by "I agree" rating on Item 3. His post-measure rating on the same item shifted to "I do not agree," and he mentioned that the procedure hardly caused any issue with his peers.

His teacher's perceptions of the intervention were positive at the beginning of the study. Upon completion of the last fading session, her ratings on the post-measure increased by 1 point for eight items and decreased for one item. Her rating for Item 6 "Most teachers would find this intervention suitable for the behavior problem described" shifted from "Strongly Agree" to "Agree." Regarding this change, she commented that she was concerned about the duration of the intervention as she believed it was a bit time-consuming for the general education classroom. Nevertheless, her rating on the post-measure indicated that she was satisfied with the outcomes, found the intervention acceptable, and considered the goals socially valid.

## Discussion

The purpose of this study was to investigate the effectiveness of self-management for two Grade 2 students diagnosed with ASD and ADHD in general education Australian primary schools. The results support the hypotheses that the implementation of self-management would result in improved compliance, be associated with a concomitant improvement in on-task behavior, that these changes would maintain following the fading of the intervention, and both participants and their teachers would consider self-management socially valid. These findings are consistent with the results of a previous study on increasing compliance and on-task behavior in a student with Asperger's syndrome using a self-management intervention (Wilkinson 2008).

The degree of improvement in compliance was socially significant, in line with normative expectations of between 60 and 90%, as suggested by McMahon and Forehand (2003). Furthermore, effect size calculations suggest that self-management met the required standard to be classified as *very effective* for both participants in increasing the primary target behavior, compliance, in all settings. This finding should be treated with caution because with one participant, Sean, data available for one setting, English support, fails to meet the minimum criterion of three data points per phase set by the What Works Clearinghouse (Kratochwill et al. 2010).

In addition to the increase in compliance, the intervention was associated with concomitant improvements in on-task behavior, such that the students' on-task

behavior was comparable to, or better than, that of their peers. The high levels of compliance and on-task behavior were maintained at follow-up across all settings for both participants.

In addition to being highly effective, the intervention was also deemed acceptable by both teachers and students. Even though both teachers and students were positive about the intervention before the study commenced, their ratings of the intervention further increased by the end of the study.

### Limitations

Several considerations are relevant in the extrapolation of these findings to other situations. First, the usual limitations regarding the external validity of the findings apply. Systematic replication is, therefore, recommended.

Secondly, a potential confound occurred in Sean's case during the intervention phase in English where following two intervention data points and a short vacation break, an additional teacher was introduced to the classroom. Though there was an immediate experimental effect on introduction of the intervention for Sean in this condition, the further elevated compliance rates after the break cannot be attributed to the self-management intervention alone.

A further limitation for Sean in the English support class was that only two intervention data points were able to be collected. Unfortunately, such limitations are not uncommon in applied work of this nature (see, for example, Roberts et al. 2019).

### Theoretical Implication

The approach to addressing non-compliance in this study was based on a skill-deficit paradigm, where rather than aiming to reduce non-compliance we sought to increase rates of compliance. This is in line with Skinner's analysis of verbal behavior where compliance can be seen as a listener skill (Cooper et al. 2007). The results support the notion that compliance is a skill that can be taught. This has theoretical and practical implications. Researchers and therapists might focus more on increasing compliance rather than seeking to reduce non-compliant behaviors.

### Future Direction

While the results from this study suggest that self-management is an effective and socially valid way of increasing compliance in junior elementary students with ASD, the social validity of the procedure could be enhanced. One way of doing this could be by incorporating wearable technology to facilitate self-recording. Sean expressed his concern numerous times of not wanting to appear different from his peers and was initially hesitant about using the paper and pencil-based self-monitoring tool during the intervention phase. Paper and pencil-based self-monitoring also has limitations in classroom settings that are more dynamic and interactive (where children move around the classroom, or engage in activities that are not table-based). Wearable technology (such as smartwatches) in conjunction with

appropriate self-monitoring apps could overcome these limitations. Coupled with some kind of prompting mechanism, they could also facilitate greater independence from the researcher. The self-recording process would be more discrete. Busacca et al. (2016) have recently piloted the use of such technology in the classroom, and behavioral improvements and high social validity were observed. Future research exploring the effects and the acceptability of such technology in self-management is clearly warranted.

## Conclusion

Compliance is regarded as a critical skill for academic and social success at school (Ducharme and Shecter 2011). Not only did this study demonstrate the acceptability and effectiveness of self-management in increasing compliance and on-task behavior in young children with ASD in primary school settings, the potential benefit of teaching self-management in order to empower the student in managing his or her own behavior, while decreasing the teacher's role in behavior management, was evident. In classrooms where teachers are often overwhelmed with instructional and classroom management demands, this intervention could play a role in alleviating teachers' stress and, in turn, fostering the inclusion of students with special educational needs in regular general education classrooms.

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